

by_normal_form_q11 over GF(11)

July 15, 2026

The equation

The equation of the quartic curve is :

$$X_0^4 + 4X_1^4 + 4X_2^4 + 5X_0^3X_1 + 4X_0^3X_2 + 4X_0X_1^3 + 4X_1^3X_2 + 3X_0X_2^3 + 7X_1X_2^3 + 4X_0^2X_2^2 + 4X_1^2X_2^2 + 10X_0^2X_1X_2 + X_0X_1^2X_2 + X_0X_1X_2^2$$

(1, 4, 4, 5, 4, 4, 4, 3, 7, 0, 4, 4, 10, 1, 1)

$$X_0^4 + 4X_1^4 + 4X_2^4 + 5X_0^3X_1 + 4X_0^3X_2 + 4X_0X_1^3 + 4X_1^3X_2 + 3X_0X_2^3 + 7X_1X_2^3$$

General information

Number of bitangents	28
Number of points	0
Fullness	is full
Number of Kovalevski points	21
Bitangent line type (a_0, a_1, a_2)	(28, 0, 0)
Number of singular points	0

All points on the curve

The quartic has 0 points:

The points on the quartic curve are:

The points by rank are: ()

The Kovalevski points are:

$$0 : P_{125} = (3, 10, 1) = b_2 \cap b_5 \cap c_{23} \cap c_{35}$$

$$1 : P_{61} = (5, 4, 1) = a_2 \cap a_6 \cap c_{24} \cap c_{46}$$

$$2 : P_{122} = (0, 10, 1) = a_2 \cap a_3 \cap b_2 \cap b_3$$

$$3 : P_{120} = (9, 9, 1) = c_{24} \cap c_{26} \cap c_{34} \cap c_{36}$$

4 : $P_{28} = (5, 1, 1) = c_{12} \cap c_{13} \cap c_{25} \cap c_{35}$
 5 : $P_{109} = (9, 8, 1) = a_2 \cap b_6 \cap c_{15} \cap c_{34}$
 6 : $P_{26} = (3, 1, 1) = a_1 \cap a_6 \cap c_{12} \cap c_{26}$
 7 : $P_{103} = (3, 8, 1) = c_{14} \cap c_{15} \cap c_{46} \cap c_{56}$
 8 : $P_{51} = (6, 3, 1) = c_{16} \cap c_{23} \cap c_{45} \cap d$
 9 : $P_{101} = (1, 8, 1) = b_1 \cap b_3 \cap c_{15} \cap c_{35}$
 10 : $P_{11} = (8, 1, 0) = a_4 \cap a_6 \cap c_{14} \cap c_{16}$
 11 : $P_{94} = (5, 7, 1) = b_1 \cap b_5 \cap c_{16} \cap c_{56}$
 12 : $P_{21} = (8, 0, 1) = b_4 \cap b_6 \cap c_{45} \cap c_{56}$
 13 : $P_{10} = (7, 1, 0) = b_1 \cap b_6 \cap c_{13} \cap c_{36}$
 14 : $P_{78} = (0, 6, 1) = a_3 \cap a_4 \cap c_{23} \cap c_{24}$
 15 : $P_{35} = (1, 2, 1) = a_4 \cap b_5 \cap c_{13} \cap c_{26}$
 16 : $P_8 = (5, 1, 0) = a_1 \cap b_3 \cap c_{25} \cap c_{46}$
 17 : $P_{68} = (1, 5, 1) = a_1 \cap a_5 \cap c_{14} \cap c_{45}$
 18 : $P_{128} = (6, 10, 1) = a_5 \cap b_2 \cap c_{25} \cap d$
 19 : $P_{31} = (8, 1, 1) = a_5 \cap b_4 \cap c_{12} \cap c_{36}$
 20 : $P_1 = (0, 1, 0) = a_3 \cap b_4 \cap c_{34} \cap d$

The incidence structure induced by the Kovalevski points is:

Flags=(6, 16, 17, 22, 23, 26, 44, 56, 62, 73, 77, 78, 101, 102, 103, 106, 111, 115,
 135, 137, 139, 147, 149, 165, 170, 177, 184, 201, 208, 209, 210, 221, 225, 236,
 243, 244, 256, 258, 271, 277, 286, 288, 301, 304, 311, 320, 322, 324, 344, 346,
 347, 357, 365, 371, 379, 381, 392, 403, 415, 417, 423, 426, 435, 444, 446, 461,
 462, 466, 471, 486, 496, 502, 512, 516, 521, 526, 532, 541, 553, 557, 558, 575,
 585, 587)

nb_rows=28

nb_cols=21

nb_flags=84

Incma=

(0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0)
 (0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)
 (0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1)
 (0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0)
 (0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0)
 (0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)
 (0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0)
 (1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0)
 (0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0)
 (0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1)
 (1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0)
 (0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0)
 (0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0)
 (0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0)
 (0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0)
 (0, 0, 0, 0, 0, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)
 (0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0)
 (1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0)

(0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0)
(0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0)
(0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1)
(1, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)
(0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0)
(0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0)
(0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0)

The Kovalevski points sorted by rank are: (1, 8, 10, 11, 21, 26, 28, 31, 35, 51,
61, 68, 78, 94, 101, 103, 109, 120, 122, 125, 128)

The points off the curve are:

0 : $P_0 = (1, 0, 0)$	31 : $P_{31} = (8, 1, 1)$	62 : $P_{62} = (6, 4, 1)$
1 : $P_1 = (0, 1, 0)$	32 : $P_{32} = (9, 1, 1)$	63 : $P_{63} = (7, 4, 1)$
2 : $P_2 = (0, 0, 1)$	33 : $P_{33} = (10, 1, 1)$	64 : $P_{64} = (8, 4, 1)$
3 : $P_3 = (1, 1, 1)$	34 : $P_{34} = (0, 2, 1)$	65 : $P_{65} = (9, 4, 1)$
4 : $P_4 = (1, 1, 0)$	35 : $P_{35} = (1, 2, 1)$	66 : $P_{66} = (10, 4, 1)$
5 : $P_5 = (2, 1, 0)$	36 : $P_{36} = (2, 2, 1)$	67 : $P_{67} = (0, 5, 1)$
6 : $P_6 = (3, 1, 0)$	37 : $P_{37} = (3, 2, 1)$	68 : $P_{68} = (1, 5, 1)$
7 : $P_7 = (4, 1, 0)$	38 : $P_{38} = (4, 2, 1)$	69 : $P_{69} = (2, 5, 1)$
8 : $P_8 = (5, 1, 0)$	39 : $P_{39} = (5, 2, 1)$	70 : $P_{70} = (3, 5, 1)$
9 : $P_9 = (6, 1, 0)$	40 : $P_{40} = (6, 2, 1)$	71 : $P_{71} = (4, 5, 1)$
10 : $P_{10} = (7, 1, 0)$	41 : $P_{41} = (7, 2, 1)$	72 : $P_{72} = (5, 5, 1)$
11 : $P_{11} = (8, 1, 0)$	42 : $P_{42} = (8, 2, 1)$	73 : $P_{73} = (6, 5, 1)$
12 : $P_{12} = (9, 1, 0)$	43 : $P_{43} = (9, 2, 1)$	74 : $P_{74} = (7, 5, 1)$
13 : $P_{13} = (10, 1, 0)$	44 : $P_{44} = (10, 2, 1)$	75 : $P_{75} = (8, 5, 1)$
14 : $P_{14} = (1, 0, 1)$	45 : $P_{45} = (0, 3, 1)$	76 : $P_{76} = (9, 5, 1)$
15 : $P_{15} = (2, 0, 1)$	46 : $P_{46} = (1, 3, 1)$	77 : $P_{77} = (10, 5, 1)$
16 : $P_{16} = (3, 0, 1)$	47 : $P_{47} = (2, 3, 1)$	78 : $P_{78} = (0, 6, 1)$
17 : $P_{17} = (4, 0, 1)$	48 : $P_{48} = (3, 3, 1)$	79 : $P_{79} = (1, 6, 1)$
18 : $P_{18} = (5, 0, 1)$	49 : $P_{49} = (4, 3, 1)$	80 : $P_{80} = (2, 6, 1)$
19 : $P_{19} = (6, 0, 1)$	50 : $P_{50} = (5, 3, 1)$	81 : $P_{81} = (3, 6, 1)$
20 : $P_{20} = (7, 0, 1)$	51 : $P_{51} = (6, 3, 1)$	82 : $P_{82} = (4, 6, 1)$
21 : $P_{21} = (8, 0, 1)$	52 : $P_{52} = (7, 3, 1)$	83 : $P_{83} = (5, 6, 1)$
22 : $P_{22} = (9, 0, 1)$	53 : $P_{53} = (8, 3, 1)$	84 : $P_{84} = (6, 6, 1)$
23 : $P_{23} = (10, 0, 1)$	54 : $P_{54} = (9, 3, 1)$	85 : $P_{85} = (7, 6, 1)$
24 : $P_{24} = (0, 1, 1)$	55 : $P_{55} = (10, 3, 1)$	86 : $P_{86} = (8, 6, 1)$
25 : $P_{25} = (2, 1, 1)$	56 : $P_{56} = (0, 4, 1)$	87 : $P_{87} = (9, 6, 1)$
26 : $P_{26} = (3, 1, 1)$	57 : $P_{57} = (1, 4, 1)$	88 : $P_{88} = (10, 6, 1)$
27 : $P_{27} = (4, 1, 1)$	58 : $P_{58} = (2, 4, 1)$	89 : $P_{89} = (0, 7, 1)$
28 : $P_{28} = (5, 1, 1)$	59 : $P_{59} = (3, 4, 1)$	90 : $P_{90} = (1, 7, 1)$
29 : $P_{29} = (6, 1, 1)$	60 : $P_{60} = (4, 4, 1)$	91 : $P_{91} = (2, 7, 1)$
30 : $P_{30} = (7, 1, 1)$	61 : $P_{61} = (5, 4, 1)$	92 : $P_{92} = (3, 7, 1)$

93 : $P_{93} = (4, 7, 1)$	107 : $P_{107} = (7, 8, 1)$	121 : $P_{121} = (10, 9, 1)$
94 : $P_{94} = (5, 7, 1)$	108 : $P_{108} = (8, 8, 1)$	122 : $P_{122} = (0, 10, 1)$
95 : $P_{95} = (6, 7, 1)$	109 : $P_{109} = (9, 8, 1)$	123 : $P_{123} = (1, 10, 1)$
96 : $P_{96} = (7, 7, 1)$	110 : $P_{110} = (10, 8, 1)$	124 : $P_{124} = (2, 10, 1)$
97 : $P_{97} = (8, 7, 1)$	111 : $P_{111} = (0, 9, 1)$	125 : $P_{125} = (3, 10, 1)$
98 : $P_{98} = (9, 7, 1)$	112 : $P_{112} = (1, 9, 1)$	126 : $P_{126} = (4, 10, 1)$
99 : $P_{99} = (10, 7, 1)$	113 : $P_{113} = (2, 9, 1)$	127 : $P_{127} = (5, 10, 1)$
100 : $P_{100} = (0, 8, 1)$	114 : $P_{114} = (3, 9, 1)$	128 : $P_{128} = (6, 10, 1)$
101 : $P_{101} = (1, 8, 1)$	115 : $P_{115} = (4, 9, 1)$	129 : $P_{129} = (7, 10, 1)$
102 : $P_{102} = (2, 8, 1)$	116 : $P_{116} = (5, 9, 1)$	130 : $P_{130} = (8, 10, 1)$
103 : $P_{103} = (3, 8, 1)$	117 : $P_{117} = (6, 9, 1)$	131 : $P_{131} = (9, 10, 1)$
104 : $P_{104} = (4, 8, 1)$	118 : $P_{118} = (7, 9, 1)$	132 : $P_{132} = (10, 10, 1)$
105 : $P_{105} = (5, 8, 1)$	119 : $P_{119} = (8, 9, 1)$	
106 : $P_{106} = (6, 8, 1)$	120 : $P_{120} = (9, 9, 1)$	

(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132)

Duals of Bitangents

dual of bitangents:

- 0 : $P_{51} = (6, 3, 1)$
- 1 : $P_{33} = (10, 1, 1)$
- 2 : $P_0 = (1, 0, 0)$
- 3 : $P_{114} = (3, 9, 1)$
- 4 : $P_{103} = (3, 8, 1)$
- 5 : $P_{76} = (9, 5, 1)$
- 6 : $P_6 = (3, 1, 0)$
- 7 : $P_{24} = (0, 1, 1)$
- 8 : $P_{25} = (2, 1, 1)$
- 9 : $P_{17} = (4, 0, 1)$
- 10 : $P_{87} = (9, 6, 1)$
- 11 : $P_{71} = (4, 5, 1)$
- 12 : $P_{122} = (0, 10, 1)$
- 13 : $P_{40} = (6, 2, 1)$
- 14 : $P_{80} = (2, 6, 1)$
- 15 : $P_{56} = (0, 4, 1)$
- 16 : $P_{42} = (8, 2, 1)$
- 17 : $P_{121} = (10, 9, 1)$
- 18 : $P_{119} = (8, 9, 1)$

$$\begin{aligned}
19 : P_5 &= (2, 1, 0) \\
20 : P_{58} &= (2, 4, 1) \\
21 : P_{19} &= (6, 0, 1) \\
22 : P_{53} &= (8, 3, 1) \\
23 : P_{99} &= (10, 7, 1) \\
24 : P_{126} &= (4, 10, 1) \\
25 : P_{92} &= (3, 7, 1) \\
26 : P_{104} &= (4, 8, 1) \\
27 : P_{22} &= (9, 0, 1)
\end{aligned}$$

Dimension of the vanishing ideal is 0

The lines and their points of contact are:

$$\begin{aligned}
a_1 &= \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 8 \end{bmatrix}_{68} \\
a_2 &= \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 10 \end{bmatrix}_{22} \\
a_3 &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{132} \\
a_4 &= \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 2 \end{bmatrix}_{98} \\
a_5 &= \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 3 \end{bmatrix}_{99} \\
a_6 &= \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 6 \end{bmatrix}_{30} \\
b_1 &= \begin{bmatrix} 1 & 8 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{107} \\
b_2 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 10 \end{bmatrix}_{10} \\
b_3 &= \begin{bmatrix} 1 & 0 & 9 \\ 0 & 1 & 10 \end{bmatrix}_{118} \\
b_4 &= \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 0 \end{bmatrix}_{84} \\
b_5 &= \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 5 \end{bmatrix}_{29} \\
b_6 &= \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 6 \end{bmatrix}_{90} \\
c_{12} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}_1 \\
c_{13} &= \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 9 \end{bmatrix}_{69} \\
c_{14} &= \begin{bmatrix} 1 & 0 & 9 \\ 0 & 1 & 5 \end{bmatrix}_{113}
\end{aligned}$$

$$\begin{aligned}
c_{15} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 7 \end{bmatrix}_7 \\
c_{16} &= \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 9 \end{bmatrix}_{45} \\
c_{23} &= \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}_{14} \\
c_{24} &= \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \end{bmatrix}_{38} \\
c_{25} &= \begin{bmatrix} 1 & 9 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{119} \\
c_{26} &= \begin{bmatrix} 1 & 0 & 9 \\ 0 & 1 & 7 \end{bmatrix}_{115} \\
c_{34} &= \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 0 \end{bmatrix}_{60} \\
c_{35} &= \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 8 \end{bmatrix}_{44} \\
c_{36} &= \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 4 \end{bmatrix}_{16} \\
c_{45} &= \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 1 \end{bmatrix}_{85} \\
c_{46} &= \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 4 \end{bmatrix}_{100} \\
c_{56} &= \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 3 \end{bmatrix}_{87} \\
d &= \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \end{bmatrix}_{24}
\end{aligned}$$

Rank of lines: (68, 22, 132, 98, 99, 30, 107, 10, 118, 84, 29, 90, 1, 69, 113, 7, 45, 14, 38, 119, 115, 60, 44, 16, 85, 100, 87, 24)

Line type: 0^{28}

28	0	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 0
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point types:

point types for points off the curve: 4^{21} , 3^{84} , 0^{28}

21	4	125, 61, 122, 120, 28, 109, 26, 103, 51, 101, 11, 94, 21, 10, 78, 35, 8, 68, 128, 31, 1
84	3	30, 14, 29, 116, 55, 27, 6, 121, 123, 59, 49, 47, 2, 119, 113, 22, 91, 45, 95, 97, 100, 24, 12, 52, 53, 60, 111, 56, 44, 117, 43, 4, 126, 42, 84, 20, 40, 80, 19, 62, 18, 37, 130, 73, 17, 131, 75, 76, 77, 38, 129, 70, 69, 82, 34, 33, 85, 86, 87, 88, 89, 67, 0, 92, 65, 32, 64, 96, 63, 98, 127, 3, 132, 102, 25, 104, 105, 106, 107, 108, 54, 110, 124, 112
28	0	58, 46, 15, 115, 74, 99, 83, 71, 114, 118, 16, 36, 90, 50, 48, 41, 79, 72, 23, 5, 57, 13, 7, 66, 93, 9, 39, 81

Lines on points off the curve:

Off point $0 = P_0 = (1, 0, 0)$ lies on 3 bisecants : $\{ 7, 12, 15 \}$
Off point $1 = P_1 = (0, 1, 0)$ lies on 4 bisecants : $\{ 2, 9, 21, 27 \}$
Off point $2 = P_2 = (0, 0, 1)$ lies on 3 bisecants : $\{ 2, 6, 19 \}$
Off point $3 = P_3 = (1, 1, 1)$ lies on 3 bisecants : $\{ 12, 16, 25 \}$
Off point $4 = P_4 = (1, 1, 0)$ lies on 3 bisecants : $\{ 1, 4, 22 \}$
Off point $5 = P_5 = (2, 1, 0)$ lies on 0 bisecants : $\{ \}$
Off point $6 = P_6 = (3, 1, 0)$ lies on 3 bisecants : $\{ 10, 18, 24 \}$
Off point $7 = P_7 = (4, 1, 0)$ lies on 0 bisecants : $\{ \}$
Off point $8 = P_8 = (5, 1, 0)$ lies on 4 bisecants : $\{ 0, 8, 19, 25 \}$
Off point $9 = P_9 = (6, 1, 0)$ lies on 0 bisecants : $\{ \}$
Off point $10 = P_{10} = (7, 1, 0)$ lies on 4 bisecants : $\{ 6, 11, 13, 23 \}$
Off point $11 = P_{11} = (8, 1, 0)$ lies on 4 bisecants : $\{ 3, 5, 14, 16 \}$
Off point $12 = P_{12} = (9, 1, 0)$ lies on 3 bisecants : $\{ 17, 20, 26 \}$
Off point $13 = P_{13} = (10, 1, 0)$ lies on 0 bisecants : $\{ \}$
Off point $14 = P_{14} = (1, 0, 1)$ lies on 3 bisecants : $\{ 1, 17, 23 \}$
Off point $15 = P_{15} = (2, 0, 1)$ lies on 0 bisecants : $\{ \}$
Off point $16 = P_{16} = (3, 0, 1)$ lies on 0 bisecants : $\{ \}$
Off point $17 = P_{17} = (4, 0, 1)$ lies on 3 bisecants : $\{ 16, 18, 22 \}$
Off point $18 = P_{18} = (5, 0, 1)$ lies on 3 bisecants : $\{ 8, 14, 20 \}$
Off point $19 = P_{19} = (6, 0, 1)$ lies on 3 bisecants : $\{ 5, 10, 27 \}$
Off point $20 = P_{20} = (7, 0, 1)$ lies on 3 bisecants : $\{ 3, 4, 25 \}$
Off point $21 = P_{21} = (8, 0, 1)$ lies on 4 bisecants : $\{ 9, 11, 24, 26 \}$
Off point $22 = P_{22} = (9, 0, 1)$ lies on 3 bisecants : $\{ 0, 13, 21 \}$
Off point $23 = P_{23} = (10, 0, 1)$ lies on 0 bisecants : $\{ \}$
Off point $24 = P_{24} = (0, 1, 1)$ lies on 3 bisecants : $\{ 2, 12, 24 \}$
Off point $25 = P_{25} = (2, 1, 1)$ lies on 3 bisecants : $\{ 1, 12, 14 \}$
Off point $26 = P_{26} = (3, 1, 1)$ lies on 4 bisecants : $\{ 0, 5, 12, 20 \}$

Off point 27 = $P_{27} = (4, 1, 1)$ lies on 3 bisecants : $\{ 3, 11, 12 \}$
 Off point 28 = $P_{28} = (5, 1, 1)$ lies on 4 bisecants : $\{ 12, 13, 19, 22 \}$
 Off point 29 = $P_{29} = (6, 1, 1)$ lies on 3 bisecants : $\{ 12, 26, 27 \}$
 Off point 30 = $P_{30} = (7, 1, 1)$ lies on 3 bisecants : $\{ 6, 12, 18 \}$
 Off point 31 = $P_{31} = (8, 1, 1)$ lies on 4 bisecants : $\{ 4, 9, 12, 23 \}$
 Off point 32 = $P_{32} = (9, 1, 1)$ lies on 3 bisecants : $\{ 10, 12, 21 \}$
 Off point 33 = $P_{33} = (10, 1, 1)$ lies on 3 bisecants : $\{ 8, 12, 17 \}$
 Off point 34 = $P_{34} = (0, 2, 1)$ lies on 3 bisecants : $\{ 2, 5, 11 \}$
 Off point 35 = $P_{35} = (1, 2, 1)$ lies on 4 bisecants : $\{ 3, 10, 13, 20 \}$
 Off point 36 = $P_{36} = (2, 2, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 37 = $P_{37} = (3, 2, 1)$ lies on 3 bisecants : $\{ 1, 6, 24 \}$
 Off point 38 = $P_{38} = (4, 2, 1)$ lies on 3 bisecants : $\{ 8, 23, 26 \}$
 Off point 39 = $P_{39} = (5, 2, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 40 = $P_{40} = (6, 2, 1)$ lies on 3 bisecants : $\{ 22, 25, 27 \}$
 Off point 41 = $P_{41} = (7, 2, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 42 = $P_{42} = (8, 2, 1)$ lies on 3 bisecants : $\{ 0, 9, 17 \}$
 Off point 43 = $P_{43} = (9, 2, 1)$ lies on 3 bisecants : $\{ 4, 16, 21 \}$
 Off point 44 = $P_{44} = (10, 2, 1)$ lies on 3 bisecants : $\{ 14, 18, 19 \}$
 Off point 45 = $P_{45} = (0, 3, 1)$ lies on 3 bisecants : $\{ 2, 23, 25 \}$
 Off point 46 = $P_{46} = (1, 3, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 47 = $P_{47} = (2, 3, 1)$ lies on 3 bisecants : $\{ 0, 18, 26 \}$
 Off point 48 = $P_{48} = (3, 3, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 49 = $P_{49} = (4, 3, 1)$ lies on 3 bisecants : $\{ 1, 10, 19 \}$
 Off point 50 = $P_{50} = (5, 3, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 51 = $P_{51} = (6, 3, 1)$ lies on 4 bisecants : $\{ 16, 17, 24, 27 \}$
 Off point 52 = $P_{52} = (7, 3, 1)$ lies on 3 bisecants : $\{ 11, 14, 22 \}$
 Off point 53 = $P_{53} = (8, 3, 1)$ lies on 3 bisecants : $\{ 5, 9, 13 \}$
 Off point 54 = $P_{54} = (9, 3, 1)$ lies on 3 bisecants : $\{ 3, 8, 21 \}$
 Off point 55 = $P_{55} = (10, 3, 1)$ lies on 3 bisecants : $\{ 4, 6, 20 \}$
 Off point 56 = $P_{56} = (0, 4, 1)$ lies on 3 bisecants : $\{ 2, 4, 26 \}$
 Off point 57 = $P_{57} = (1, 4, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 58 = $P_{58} = (2, 4, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 59 = $P_{59} = (3, 4, 1)$ lies on 3 bisecants : $\{ 8, 11, 16 \}$
 Off point 60 = $P_{60} = (4, 4, 1)$ lies on 3 bisecants : $\{ 13, 14, 17 \}$
 Off point 61 = $P_{61} = (5, 4, 1)$ lies on 4 bisecants : $\{ 1, 5, 18, 25 \}$
 Off point 62 = $P_{62} = (6, 4, 1)$ lies on 3 bisecants : $\{ 3, 6, 27 \}$
 Off point 63 = $P_{63} = (7, 4, 1)$ lies on 3 bisecants : $\{ 0, 10, 23 \}$
 Off point 64 = $P_{64} = (8, 4, 1)$ lies on 3 bisecants : $\{ 9, 20, 22 \}$
 Off point 65 = $P_{65} = (9, 4, 1)$ lies on 3 bisecants : $\{ 19, 21, 24 \}$
 Off point 66 = $P_{66} = (10, 4, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 67 = $P_{67} = (0, 5, 1)$ lies on 3 bisecants : $\{ 2, 13, 16 \}$
 Off point 68 = $P_{68} = (1, 5, 1)$ lies on 4 bisecants : $\{ 0, 4, 14, 24 \}$
 Off point 69 = $P_{69} = (2, 5, 1)$ lies on 3 bisecants : $\{ 5, 6, 17 \}$
 Off point 70 = $P_{70} = (3, 5, 1)$ lies on 3 bisecants : $\{ 3, 19, 23 \}$
 Off point 71 = $P_{71} = (4, 5, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 72 = $P_{72} = (5, 5, 1)$ lies on 0 bisecants : $\{ \}$

Off point 73 = $P_{73} = (6, 5, 1)$ lies on 3 bisecants : { 1, 20, 27 }
 Off point 74 = $P_{74} = (7, 5, 1)$ lies on 0 bisecants : { }
 Off point 75 = $P_{75} = (8, 5, 1)$ lies on 3 bisecants : { 8, 9, 18 }
 Off point 76 = $P_{76} = (9, 5, 1)$ lies on 3 bisecants : { 21, 22, 26 }
 Off point 77 = $P_{77} = (10, 5, 1)$ lies on 3 bisecants : { 10, 11, 25 }
 Off point 78 = $P_{78} = (0, 6, 1)$ lies on 4 bisecants : { 2, 3, 17, 18 }
 Off point 79 = $P_{79} = (1, 6, 1)$ lies on 0 bisecants : { }
 Off point 80 = $P_{80} = (2, 6, 1)$ lies on 3 bisecants : { 4, 8, 10 }
 Off point 81 = $P_{81} = (3, 6, 1)$ lies on 0 bisecants : { }
 Off point 82 = $P_{82} = (4, 6, 1)$ lies on 3 bisecants : { 20, 24, 25 }
 Off point 83 = $P_{83} = (5, 6, 1)$ lies on 0 bisecants : { }
 Off point 84 = $P_{84} = (6, 6, 1)$ lies on 3 bisecants : { 0, 11, 27 }
 Off point 85 = $P_{85} = (7, 6, 1)$ lies on 3 bisecants : { 1, 13, 26 }
 Off point 86 = $P_{86} = (8, 6, 1)$ lies on 3 bisecants : { 9, 16, 19 }
 Off point 87 = $P_{87} = (9, 6, 1)$ lies on 3 bisecants : { 6, 14, 21 }
 Off point 88 = $P_{88} = (10, 6, 1)$ lies on 3 bisecants : { 5, 22, 23 }
 Off point 89 = $P_{89} = (0, 7, 1)$ lies on 3 bisecants : { 0, 2, 22 }
 Off point 90 = $P_{90} = (1, 7, 1)$ lies on 0 bisecants : { }
 Off point 91 = $P_{91} = (2, 7, 1)$ lies on 3 bisecants : { 11, 19, 20 }
 Off point 92 = $P_{92} = (3, 7, 1)$ lies on 3 bisecants : { 4, 13, 18 }
 Off point 93 = $P_{93} = (4, 7, 1)$ lies on 0 bisecants : { }
 Off point 94 = $P_{94} = (5, 7, 1)$ lies on 4 bisecants : { 6, 10, 16, 26 }
 Off point 95 = $P_{95} = (6, 7, 1)$ lies on 3 bisecants : { 14, 23, 27 }
 Off point 96 = $P_{96} = (7, 7, 1)$ lies on 3 bisecants : { 5, 8, 24 }
 Off point 97 = $P_{97} = (8, 7, 1)$ lies on 3 bisecants : { 1, 3, 9 }
 Off point 98 = $P_{98} = (9, 7, 1)$ lies on 3 bisecants : { 17, 21, 25 }
 Off point 99 = $P_{99} = (10, 7, 1)$ lies on 0 bisecants : { }
 Off point 100 = $P_{100} = (0, 8, 1)$ lies on 3 bisecants : { 2, 15, 20 }
 Off point 101 = $P_{101} = (1, 8, 1)$ lies on 4 bisecants : { 6, 8, 15, 22 }
 Off point 102 = $P_{102} = (2, 8, 1)$ lies on 3 bisecants : { 15, 16, 23 }
 Off point 103 = $P_{103} = (3, 8, 1)$ lies on 4 bisecants : { 14, 15, 25, 26 }
 Off point 104 = $P_{104} = (4, 8, 1)$ lies on 3 bisecants : { 4, 5, 15 }
 Off point 105 = $P_{105} = (5, 8, 1)$ lies on 3 bisecants : { 0, 3, 15 }
 Off point 106 = $P_{106} = (6, 8, 1)$ lies on 3 bisecants : { 15, 18, 27 }
 Off point 107 = $P_{107} = (7, 8, 1)$ lies on 3 bisecants : { 15, 17, 19 }
 Off point 108 = $P_{108} = (8, 8, 1)$ lies on 3 bisecants : { 9, 10, 15 }
 Off point 109 = $P_{109} = (9, 8, 1)$ lies on 4 bisecants : { 1, 11, 15, 21 }
 Off point 110 = $P_{110} = (10, 8, 1)$ lies on 3 bisecants : { 13, 15, 24 }
 Off point 111 = $P_{111} = (0, 9, 1)$ lies on 3 bisecants : { 2, 10, 14 }
 Off point 112 = $P_{112} = (1, 9, 1)$ lies on 3 bisecants : { 5, 19, 26 }
 Off point 113 = $P_{113} = (2, 9, 1)$ lies on 3 bisecants : { 3, 22, 24 }
 Off point 114 = $P_{114} = (3, 9, 1)$ lies on 0 bisecants : { }
 Off point 115 = $P_{115} = (4, 9, 1)$ lies on 0 bisecants : { }
 Off point 116 = $P_{116} = (5, 9, 1)$ lies on 3 bisecants : { 4, 11, 17 }
 Off point 117 = $P_{117} = (6, 9, 1)$ lies on 3 bisecants : { 8, 13, 27 }
 Off point 118 = $P_{118} = (7, 9, 1)$ lies on 0 bisecants : { }

Off point 119 = $P_{119} = (8, 9, 1)$ lies on 3 bisecants : $\{ 6, 9, 25 \}$
 Off point 120 = $P_{120} = (9, 9, 1)$ lies on 4 bisecants : $\{ 18, 20, 21, 23 \}$
 Off point 121 = $P_{121} = (10, 9, 1)$ lies on 3 bisecants : $\{ 0, 1, 16 \}$
 Off point 122 = $P_{122} = (0, 10, 1)$ lies on 4 bisecants : $\{ 1, 2, 7, 8 \}$
 Off point 123 = $P_{123} = (1, 10, 1)$ lies on 3 bisecants : $\{ 7, 11, 18 \}$
 Off point 124 = $P_{124} = (2, 10, 1)$ lies on 3 bisecants : $\{ 7, 13, 25 \}$
 Off point 125 = $P_{125} = (3, 10, 1)$ lies on 4 bisecants : $\{ 7, 10, 17, 22 \}$
 Off point 126 = $P_{126} = (4, 10, 1)$ lies on 3 bisecants : $\{ 0, 6, 7 \}$
 Off point 127 = $P_{127} = (5, 10, 1)$ lies on 3 bisecants : $\{ 7, 23, 24 \}$
 Off point 128 = $P_{128} = (6, 10, 1)$ lies on 4 bisecants : $\{ 4, 7, 19, 27 \}$
 Off point 129 = $P_{129} = (7, 10, 1)$ lies on 3 bisecants : $\{ 7, 16, 20 \}$
 Off point 130 = $P_{130} = (8, 10, 1)$ lies on 3 bisecants : $\{ 7, 9, 14 \}$
 Off point 131 = $P_{131} = (9, 10, 1)$ lies on 3 bisecants : $\{ 5, 7, 21 \}$
 Off point 132 = $P_{132} = (10, 10, 1)$ lies on 3 bisecants : $\{ 3, 7, 26 \}$

The gradient

The gradient of the quartic curve is :

$$4X_0^3 + 4X_1^3 + 3X_2^3 + 4X_0^2X_1 + X_0^2X_2 + X_1^2X_2 + 8X_0X_2^2 + X_1X_2^2 + 9X_0X_1X_2$$

$$(4, 4, 3, 4, 1, 0, 1, 8, 1, 9)$$

$$5X_0^3 + 5X_1^3 + 7X_2^3 + 10X_0^2X_2 + X_0X_1^2 + X_1^2X_2 + X_0X_2^2 + 8X_1X_2^2 + 2X_0X_1X_2$$

$$(5, 5, 7, 0, 10, 1, 1, 1, 8, 2)$$

$$4X_0^3 + 4X_1^3 + 5X_2^3 + 10X_0^2X_1 + 8X_0^2X_2 + X_0X_1^2 + 8X_1^2X_2 + 9X_0X_2^2 + 10X_1X_2^2 + 2X_0X_1X_2$$

$$(4, 4, 5, 10, 8, 1, 8, 9, 10, 2)$$